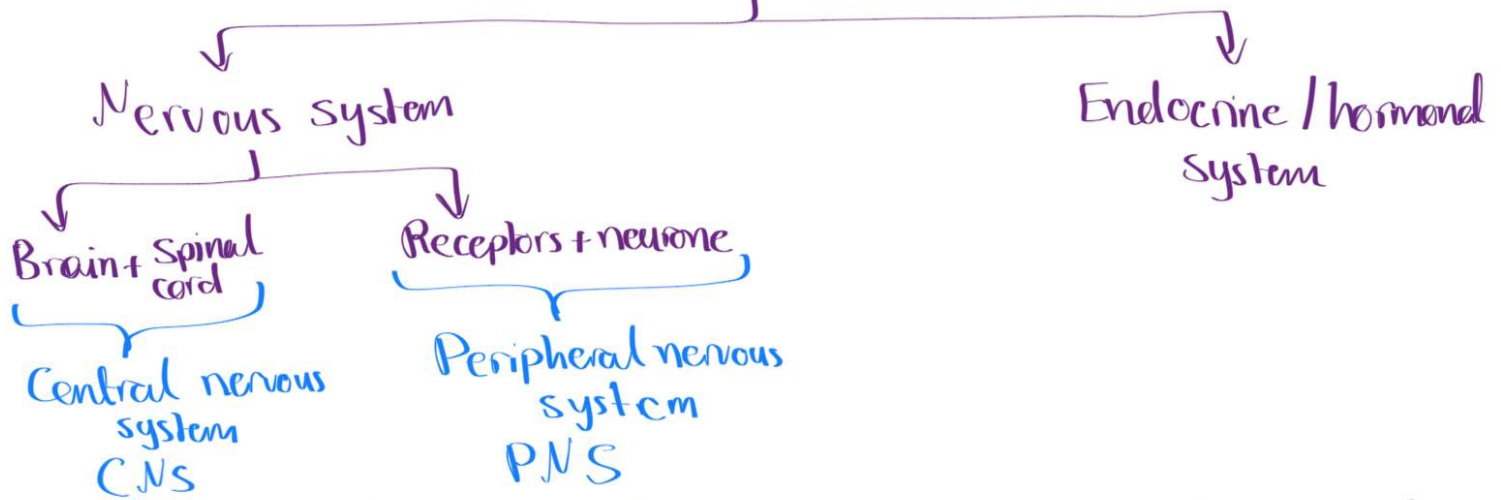


Coordination
and
response

Coordination & response



Nervous system has the role of coordinating and regulating of body functions.

Coordination: The way all the organs and system of the body are made to work efficiently together.

Stimuli: Changes in the internal or external environment of an organism

Receptors: Sensitive cells that detect stimuli

Sense organs: A group of receptor cells responding to specific stimuli: (light, sound, touch, temperature, chemicals)

Reflex action: Automatically and rapidly integrating and coordinating stimuli with responses of effectors

Reflex arc: Pathway of reflex action

Stimuli → receptors → Sensory neurone → Spinal cord / relay neurone

↳ Functions:

1- Detect stimuli

2- Send electric impulses

effectors ← motor neurone

↳ Examples:

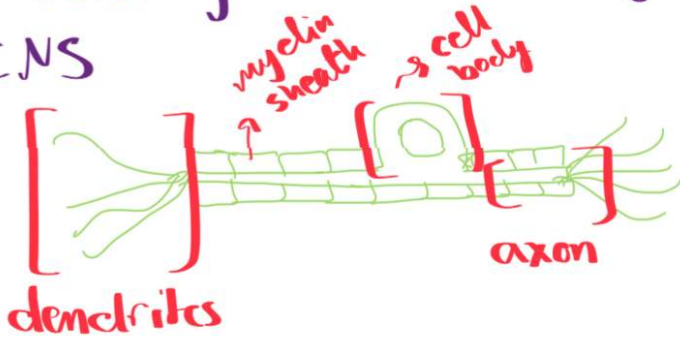
1- Muscles

2- Glands

3 Types of neurones:

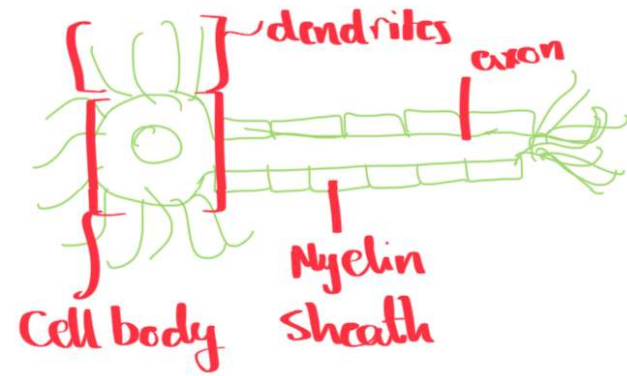
1- Sensory neurones: Carry impulses from receptors to CNS

- * Cell body in the middle
- * long dendrites, short axons.

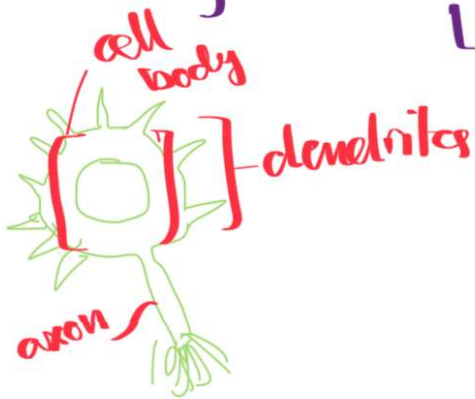


2- Motor neurones: Carry impulse from CNS to effectors.

- * Cell body at one end
- * Long axon, short dendrite



3- Relay neurone: Connects sensory with motor neurone
↳ Brain & spinal cord.



- * Short axon & dendrite.
- * No myelin sheath

Direction of impulse:

- * Sensory → relay → motor
- * Dendrite → cell body → axon

Impulse: Electrical signals that pass along nerve cells.

Example: effectors: muscles in the upper arm

→ You touch something hot, so:

1- bicep contracts → tricep relax → so you bend your arm at the elbow

2- tricep contracts → bicep relax → so you straighten your arm

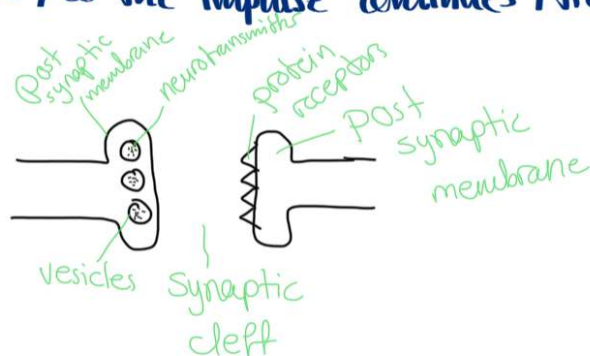
Antagonistic muscles: Muscles that work in pairs by working opposite to each other.

Synapse: A junction between two neurones

→ **Function:** Ensure the impulse travels in one direction only

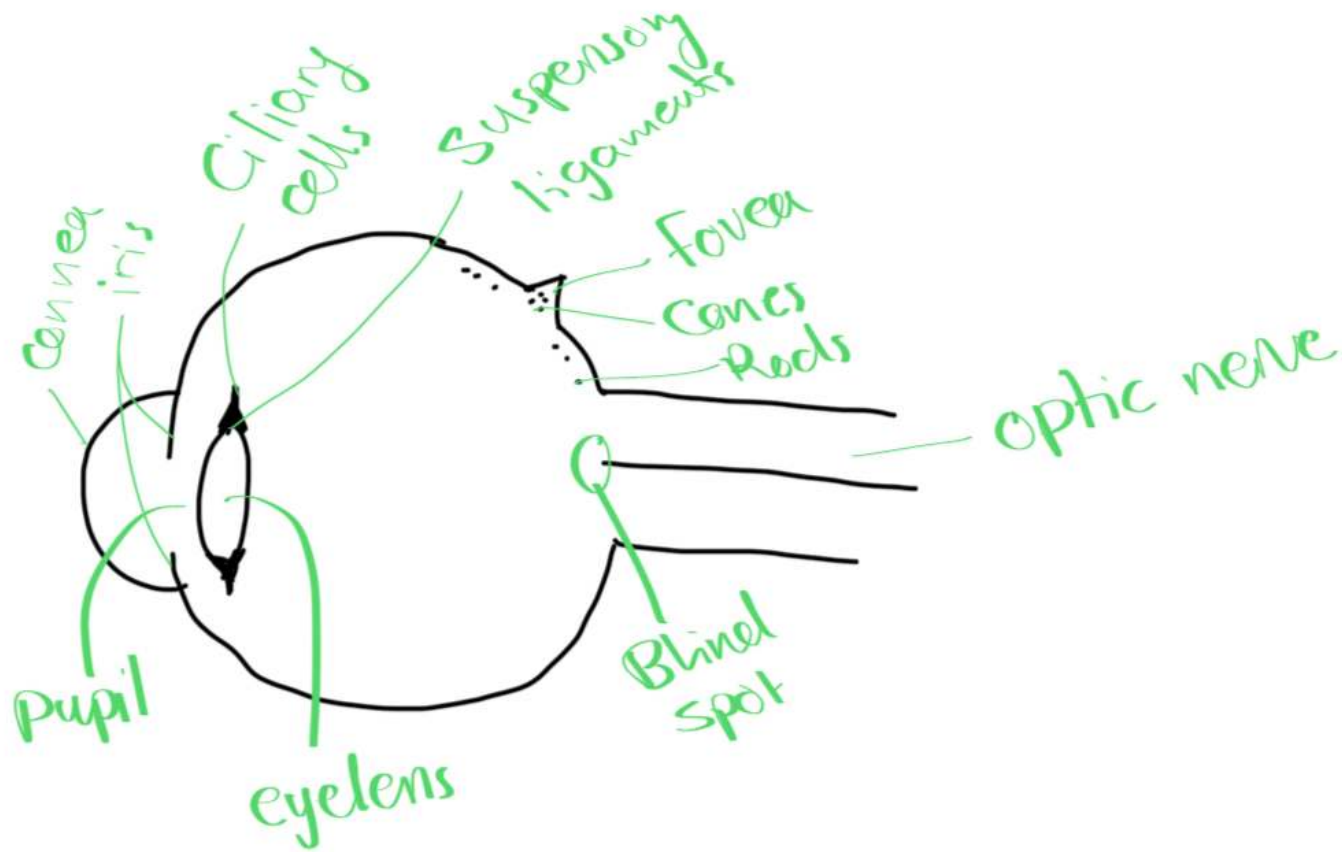
How does an impulse pass through synapse?

As the impulse arrives at the end of the neuron/axon, vesicles move and fuse with the pre-synaptic membrane, neurotransmitters are released from vesicles and diffuse across the synaptic gap. then neurotransmitters bind complementary with protein receptors on post-synaptic membrane, so the impulse continues / travels to other neurone



Why does the impulse travel in one direction only?

Because vesicles which contain neurotransmitters are found only at one side of the synapse, and the protein receptors are found at the other end of the synapse.



Cornea: Refracts the light

Iris: Controls how much light enters the pupil

lens: Focuses the light on the retina

Retina: Contains light receptors, some sensitive to light of different colors

Optic nerve: Carries impulses to the brain

Blind spot: No light receptors

Pupil reflex:

In bright light: Pupil gets smaller in size to protect the retina from damage. This happens when the circular muscles of iris contract, and the radial muscles of iris relax.

In dim light: Pupil gets bigger in size to let more light pass through to the retina. This happens when the circular muscles of iris relax, and the radial muscles of iris contract.

Accommodation

For close objects: The lens gets shorter and thicker (more convex), so suspensory ligaments slacken, and ciliary muscles contract.

For distant objects: The lens gets longer and thinner (less convex), so suspensory ligaments contract, and ciliary muscles slacken.

* Cones and rods are sensitive cells found on the retina. They are sensitive to light. They convert light to electrical impulses and send it to the brain through the optic nerve.

* Cones are concentrated in the middle of the retina, called fovea, and some of them are present in the margin of the retina, and they give colored images. Cones are sensitive to bright light. Three types of cones:

- * Red

- * Green

- * Blue

* Rods are found on the margin of the retina, they give black and white images. Rods work in the dim light.

Hormone: A chemical substance secreted from glands and travels through blood to alter target organs

Examples:

- **Insulin:** Secreted from pancreas. Has the function of decreasing the blood glucose concentration.
- **Oestrogen:** Secreted from ovaries. Has the function of controlling the female secondary sexual characteristics.
- **Testosterone:** Secreted from testis. Has the function of controlling the male secondary sexual characteristics.
- **Adrenaline:** It's the fight or flight hormone. Secreted from the adrenal gland.

Functions:

- * Increase the heart rate & blood pressure
- * Increasing breathing rate
- * Increase blood glucose concentration
- * Increase sweating
- * Dilate pupils.

Differences between endocrine and nervous systems

Nervous	Endocrine
* Faster response	* Slower response
* Short lasting effect	* Long lasting effect
* Electric impulse	* Chemical impulse
* Impulse travels through neurone	* Impulse travels through blood

Sensitivity in plants: - Light
- Water
- Gravity } Stimuli.

Gravitropism: A response in which parts of a plant grow towards or away from gravity

Phototropism: A response in which parts of a plant grow towards or away from the direction of light.

→ Plants show response by bending away or toward stimuli.

Why do plants bend towards light?

* To get more light, to do more photosynthesis, to produce more glucose and grow more

Phototropism

Positive:
by the shoot;
plants grow towards
the light

Negative:
by the roots;
plants grow away
from the light.

How does the plant bend toward light?

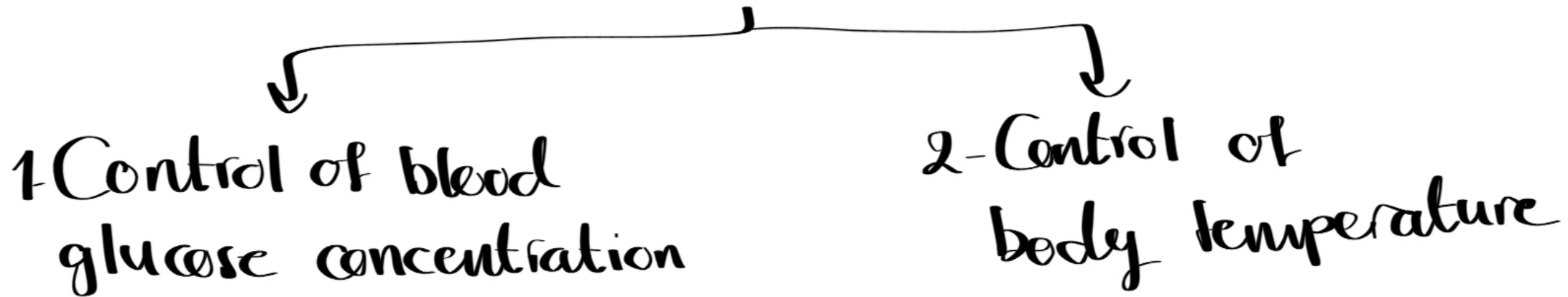
Auxin, which is the plant growth hormone, is made at the tip of the shoot. Auxin is sensitive to light, so it moves towards the shaded part of shoot by diffusion, so there will be unequal distribution of auxin. Auxin stimulates cell elongation, so uneven growth at one side of shoot, causing the shoot to bend towards light.

X-Roots show positive gravitropism, they grow downwards. why?

1-To get more water and minerals

2-To anchor the plant.

Homeostasis: Maintenance of a constant internal environment by negative feedback



1- Control of blood glucose concentration:

* After eating a meal that is rich with carbohydrates, blood glucose concentration increases. This is detected by receptors in the pancreas. Pancreas secretes a hormone called insulin, which travels in the blood to the liver. Insulin stimulates the liver to respire more aerobically, consuming more glucose, so the glucose concentration in blood decreases back to normal. Insulin stimulates the liver to convert excess glucose to glycogen, which is then stored in the liver.

* After fasting for a long time, blood glucose concentration decreases. This decrease in blood glucose concentration is detected by receptors in the pancreas. Pancreas secretes a hormone called glucagon, which travels in the blood to the liver. Glucagon stimulates the liver to break glycogen into glucose, then the glucose is released into the blood. So the blood glucose increases back to normal.

- * Insulin and glucagon are both hormones
- * Both are secreted from pancreas
- * Both are made of protein

Type one diabetes: Happens when the islet cells in the pancreas do not release enough insulin

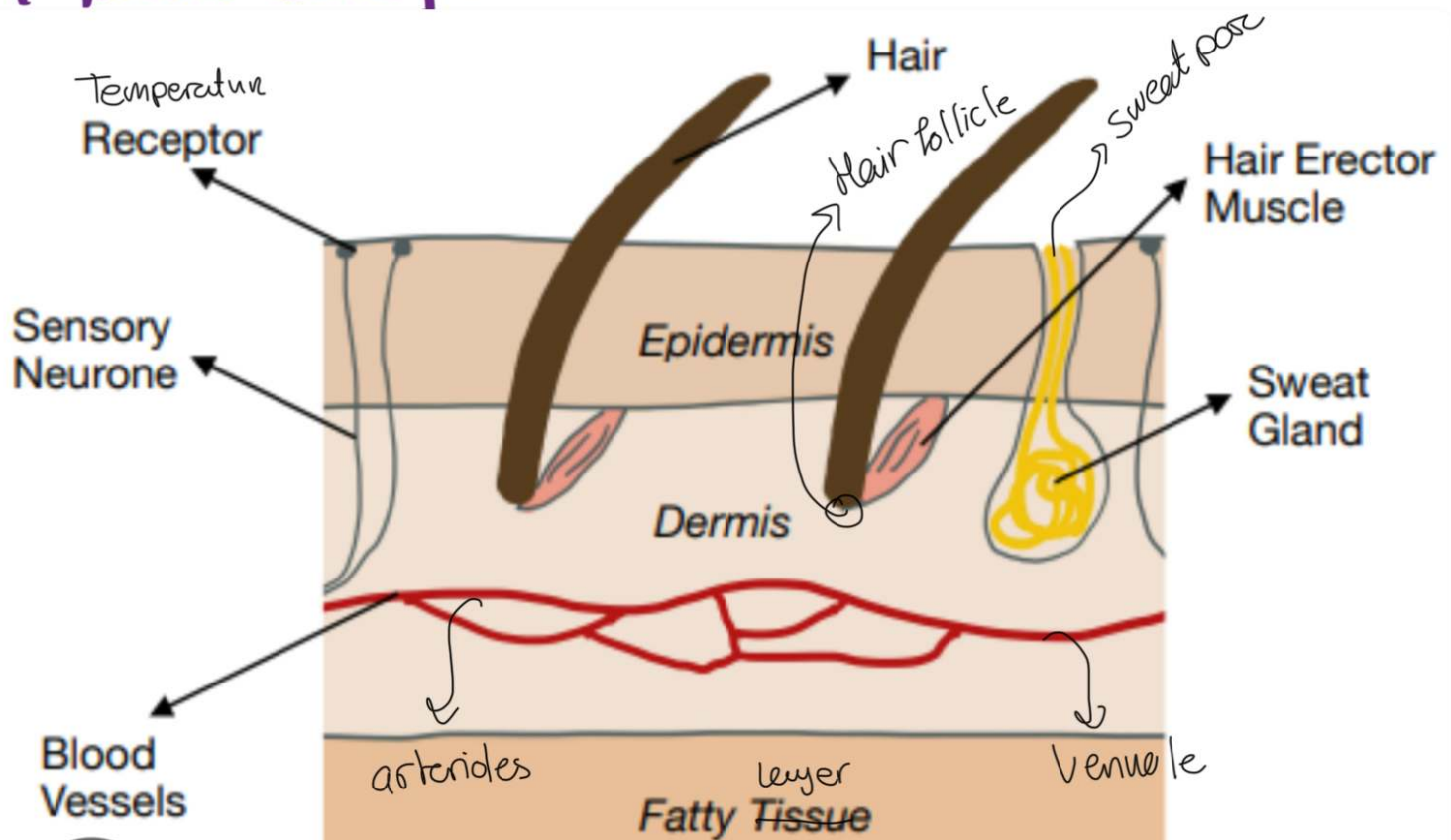
Treatment of diabetes:

- 1- Insulin injection
- 2- Regular blood glucose concentration checkup
- 3- Regular exercise
- 4- Balance diet with more fruits and vegetables

2- Maintaining constant body temperature:

Reasons to maintain body temperature:

- 1- Avoid damage to the cell membrane
- 2- Enzymes do not denature
- 3- Maintaining a constant rate of temperature
- 4- Avoid dehydration



* At high temperature, temperature receptor detects the high temperature and send an electric impulse through sensory neurone to the thermoregulatory center-hypothalamus in the brain. The hypothalamus send impulse through motor neurone to the sweat gland, so more sweating. Sweat evaporates from the surface of the skin, taking the heat from the body, so the body temperature falls. Hair erector muscle relaxes, so hair lays down, so heat is lost to the surroundings by radiation. Vasodilation of arterioles, and shunt vessels constrict, so more blood flows through capillaries so heat is lost to the surroundings by convection or radiation.

* At low temperature, temperature receptors detects decrease in temperature, and send electric impulse through sensory neurone to the hypothalamus in the brain. Hypothalamus send impulse through motor neurone to sweat gland decreasing sweating, so no heat is lost by evaporation of sweat. Hair erector muscle contracts, so the hair stands up, trapping a layer of air, which acts as an insulating layer, so heat is not lost to the surroundings. Vasoconstriction of arterioles, dilation of shunt vessels, so less blood flows through capillaries, so heat is not lost to the surrounding by radiation or convection.

Vasodilation of arterioles: The widening of arterioles in the dermis, which allows more warm blood to flow through capillaries near the skin surface, resulting in heat loss.

Vaso constriction of arterioles: The narrowing of arterioles in the skin reduces the amount of warm blood flowing through capillaries near the surface.